ANSWER KEY & MARKING SCHEME SET 1

Chapter 5: **Life Processes** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** excretion	Kidneys filter blood and form urine.
2	**(d)** CO_2, water and energy	Glucose is fully oxidised.
3	**(b)** nephron	Each kidney has about a million nephrons.
4	**(c)** transport of water	Xylem carries water and minerals.
5	**(d)** lactic acid	Lactic acid causes cramps.
6	**(d)** urinary bladder	Bladder.
7	**(b)** carbon dioxide, water, chlorophyll and sunlight	Photosynthesis needs all of these.
8	**(d)** ethanol and carbon dioxide	Fermentation.
9	**(b)** glomerulus	Glomerulus in Bowman's capsule.
10	**(c)** mitochondria	Aerobic respiration in mitochondria.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Autotrophs make their own food by photosynthesis (1); Heterotrophs depend on other organisms for food (1).	1+1
12	Long and coiled, with villi (1); Villi increase surface area and are rich in blood vessels (1).	1+1
13	Xylem transports water and minerals (1); Phloem transports food (1).	1+1
14	CO_2 from air through stomata, water from soil (1); Sunlight from the Sun and chlorophyll in leaves (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Energy is released (1); $C_6H_{12}O_6 + 6\,O_2 \longrightarrow 6\,CO_2 + 6\,H_2O + \text{energy}$ (1); Oxidation of glucose (1).	1+1+1
16	Filtration in glomerulus (1); Selective reabsorption of glucose, amino acids, water (1); Remaining waste forms urine (1).	1+1+1
17	Aerobic: CO_2 and water, more energy (1); Anaerobic: ethanol or lactic acid, less energy (1); Aerobic in mitochondria (1).	1+1+1
	OR: Water enters guard cells: they swell and the pore opens (1); When water leaves they shrink and the pore closes (1); This controls gas exchange and water loss (1).	1+1+1
18	Alveoli provide a large surface area (1); Thin walls enable gas exchange (1); Rich capillaries take oxygen and give CO_2 (1).	1+1+1
	OR: Anaerobic respiration in oxygen-poor muscles (1); Lactic acid builds up (1); Cramps relieved by rest or massage (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Chlorophyll absorbs light energy (1); Water splits into hydrogen and oxygen (1); CO_2 is reduced to carbohydrates (1); Desert plants take CO_2 at night and store it as an acid (1); Use it during the day for photosynthesis (1).	1+1+1+1+1
	OR: Kidneys, ureters, bladder, urethra (1); Filtration in glomerulus (1); Reabsorption in tubules (1); Urine flows to bladder through ureter (1); Released through urethra (1).	1+1+1+1+1
20	Mouth: salivary amylase on starch (1); Oesophagus: peristalsis (1); Stomach: HCl, pepsin, mucus (1); Small intestine: bile, pancreatic and intestinal juices (1); Large intestine absorbs water; rectum egestion (1).	1+1+1+1+1
	OR: Autotrophic: green plants (1); Parasitic: Cuscuta, tapeworm (1); Saprophytic: fungi, bread mould (1); Holozoic: Amoeba, humans (1); Heterotrophs depend on other organisms (1).	1+1+1+1+1

Mistakes to avoid:

- Writing that plants respire only at night and photosynthesise only by day; plants respire all the time and photosynthesise in the presence of light.
- Mixing up xylem and phloem: xylem carries water and minerals (one-way, upwards); phloem carries food (both directions).

- Saying pulmonary artery carries oxygenated blood; the pulmonary **artery** carries deoxygenated blood to the lungs and the pulmonary **vein** carries oxygenated blood back.
- Confusing breathing with respiration; breathing is gas exchange, respiration is the breakdown of glucose in cells to release energy.
- Stating that pyruvate is formed in the mitochondria; glucose is broken down to pyruvate in the cytoplasm.
- Believing the kidney makes urine only by filtration; selective reabsorption and the amount of water excreted are equally important steps.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 5: **Life Processes** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** ATP	ATP.
2	**(c)** dialysis	Haemodialysis.
3	**(a)** small intestine	Bile and pancreatic lipase act there.
4	**(b)** oxygen	It binds oxygen in RBCs.
5	**(b)** phloem	Translocation through phloem.
6	**(b)** its amylase breaks down starch into sugar	Salivary amylase.
7	**(c)** alveoli	Thin-walled alveoli.
8	**(a)** transpiration	Transpiration.
9	**(b)** mouth -> oesophagus -> stomach -> small intestine -> large intestine	Standard path of food.
10	**(c)** pulmonary vein	Only vein carrying oxygenated blood.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Air has more oxygen than water (1); Terrestrial organisms breathe slower and need not breathe fast (1).	1+1
12	Depends on how much excess water is in the body (1); And on the amount of dissolved waste to be excreted (1).	1+1
13	Hydrochloric acid creates an acidic medium for pepsin (1); Kills germs in food (1).	1+1
14	Aerobic: in presence of oxygen (1); Anaerobic: in absence of oxygen (yeast, muscles) (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Bile emulsifies fats and makes medium alkaline (1); Pancreatic juice: trypsin, lipase, amylase (1); Intestinal juice completes digestion (1).	1+1+1
16	Transpiration pull (1); Water is pulled through xylem (1); Root pressure helps at night (1).	1+1+1
17	Prevents mixing of oxygenated and deoxygenated blood (1); Efficient supply of oxygen (1); Needed in warm-blooded animals (1).	1+1+1
	OR: Peristaltic movements of the walls (1); Muscular contractions push food (1); Sphincter muscles regulate exit (1).	1+1+1
18	Phloem carries food from leaves (1); Uses energy (ATP) (1); Moves to roots, fruits, growing parts (1).	1+1+1
	OR: Useful substances like glucose are taken back (1); Water is reabsorbed according to need (1); Prevents loss of nutrients (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Ribs lift, diaphragm flattens: air enters (1); Air passes to alveoli (1); Oxygen enters blood by diffusion (1); CO_2 goes to the alveoli (1); Exhalation reverses the process (1).	1+1+1+1+1
	OR: Aerobic: CO_2 + water + energy in mitochondria (1); Yeast: ethanol + CO_2 + energy (1); Muscle: lactic acid + energy (1); Energy order: aerobic is the highest (1); Glucose to pyruvate in cytoplasm in all cases (1).	1+1+1+1+1
20	Four chambers (1); Deoxygenated blood from body: right atrium to right ventricle to lungs (1); Oxygenated blood: lungs to left atrium to left ventricle (1); Left ventricle pumps to the body (1); Valves prevent backflow (1).	1+1+1+1+1
	OR: Millions of alveoli: large area (1); Thin walls and capillaries (1); Oxygen carried by haemoglobin (1); CO_2 is dissolved in plasma (1); Haemoglobin has high affinity for oxygen (1).	1+1+1+1+1

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MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 5: **Life Processes** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** translocation	Phloem transport.
2	**(d)** small intestine	Bile enters the duodenum.
3	**(b)** heart	Heart pumps blood.
4	**(c)** guard cells	Guard cells.
5	**(a)** guard cells	Guard cells swell or shrink.
6	**(c)** left ventricle	Thick-walled left ventricle.
7	**(d)** gums, resins and latex	Stored in old xylem, bark, leaves.
8	**(d)** small intestine	Villi increase surface area.
9	**(b)** prevent backflow of blood	Ensure one-way flow.
10	**(a)** chlorophyll	Chlorophyll.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Allows a highly efficient supply of oxygen (1); Required by warm-blooded animals for high energy needs (1).	1+1
12	Break complex food molecules into simple ones (1); Work as biological catalysts (1).	1+1
13	Oxygen is carried by haemoglobin (1); CO_2 is dissolved mostly in plasma (1).	1+1
14	Waste stored in cell vacuoles/leaves that fall off (1); Excreted as gums and resins from old xylem (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Blood passes twice through the heart in one cycle (1); Pulmonary circulation: heart to lungs and back (1); Systemic circulation: heart to body and back (1).	1+1+1
16	Peristalsis moves food (1); Villi increase absorptive area (1); Long intestine gives time for digestion (1).	1+1+1
17	Raw materials: CO_2, water, sunlight, chlorophyll (1); $6\,CO_2 + 12\,H_2O \longrightarrow C_6H_{12}O_6 + 6\,O_2 + 6\,H_2O$ (1); Products: glucose and oxygen (1).	1+1+1
	OR: Arteries carry blood from heart; veins to the heart (1); Arteries have thick elastic walls; veins thin (1); Veins have valves (1).	1+1+1
18	Pseudopodia engulf food particle (1); Food vacuole is formed (1); Digestion within vacuole and waste is expelled (1).	1+1+1
	OR: Take in water through mouth (1); Pass over gills with rich blood supply (1); Dissolved oxygen is absorbed (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Roots absorb water from soil (1); Xylem tubes carry water upward (1); Transpiration creates suction (1); Root pressure helps at night (1); Cohesion keeps column of water continuous (1).	1+1+1+1+1
	OR: Salivary amylase: starch to sugar (1); Pepsin: proteins to peptides (1); Bile: emulsification (1); Trypsin and lipase in small intestine (1); Final products: glucose, amino acids, fatty acids (1).	1+1+1+1+1
20	Phloem transports sugars (1); Uses ATP, so is active (1); Bidirectional movement (1); Xylem is unidirectional and passive (1); Sieve tubes and companion cells (1).	1+1+1+1+1
	OR: Plants need less energy (1); RBC: haemoglobin carries oxygen (1); WBC: fight infection (1); Platelets: clotting (1); Plasma carries food, CO_2, waste (1).	1+1+1+1+1

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